

Polarity-Aware Logic Optimization with Mixed Majority-Minority Inverter Graphs

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Abstract—Majority-inverter graphs (MIGs) encode signal polarity on complemented edges. Explicit minority (MIN) nodes preserve Boolean expressiveness but change the optimization representation. We evaluate a two-stage native MAJ/MIN flow against a pure-MIG flow with comparable generic structural optimization from the same starting networks. Across twelve EPFL circuits, both flows pass raw ABC combinational equivalence checking against the original functions. The mMIG flow reduces aggregate MAJ+MIN nodes by 0.89% (15,205 to 15,070) and nonconstant complemented graph connections by 7.38% (6,501 to 6,021); the latter improve on eleven circuits and tie on one. A round-capped pure-MIG search retains a 7.43% edge gap, although two circuits do not reach the mMIG runtime. Direct MIN lowering followed by pure-MIG phase optimization yields 6,668 complemented connections, 2.57% more than the matched pure MIG. The measured reduction is specific to the native mixed representation and does not establish mapped area or inverter-cell savings.

Index Terms—logic synthesis, majority-inverter graphs, minority gates, polarity optimization

I. INTRODUCTION

Majority-inverter graphs (MIGs) represent combinational logic with three-input majority nodes and complemented edges [1]. The complement bit on a connection records signal polarity. Pure-MIG polarity optimization can move these bits without changing the graph topology [2]. A mixed majority-minority inverter graph (mMIG) additionally labels nodes as minority (MIN). Because MIN is the complement of MAJ, this changes neither Boolean expressiveness nor the number of inputs to a node. It does change the available native polarity assignments and mixed-graph rewrite choices [3].

The question addressed here is whether a native MAJ/MIN flow finds a different structural outcome than a pure-MIG flow that also performs structural optimization. A comparison with only fixed-topology phase optimization cannot separate the value of the mixed representation from generic rewriting and resubstitution effort. We therefore run both flows from the same archived MIG for each circuit and report logic nodes and complemented graph connections separately.

This letter contributes (i) a two-stage mMIG flow combining mixed structural optimization with fixed-topology phase reassignment, (ii) a matched pure-MIG structural and phase control, and (iii) twelve-circuit results including

raw equivalence checks, direct MIN lowering, and a pure-MIG time-budget sensitivity test.¹ The mMIG flow has 0.89% fewer aggregate logic nodes and 7.38% fewer native complemented connections than the matched control, while taking about ten times longer. These data describe the tested structural representations; they do not isolate MIN from every other implementation difference.

II. MIXED MAJ/MIN REPRESENTATION

For Boolean inputs a, b, c , the node functions are

$$\begin{aligned} \text{MAJ}(a, b, c) &= ab + ac + bc, \\ \text{MIN}(a, b, c) &= \overline{\text{MAJ}(a, b, c)}. \end{aligned} \quad (1)$$

An mMIG is a directed acyclic graph with MAJ or MIN internal nodes and regular or complemented connections. A MIN node can be represented by MAJ plus output complementation, so a mixed graph does not add Boolean functions. It makes that polarity an explicit node choice, which permits local gate-type and edge-phase changes during optimization.

We report $G = N_{\text{MAJ}} + N_{\text{MIN}}$ and I , the number of complemented connections whose source is nonconstant. I includes gate-input and primary-output connections but excludes connections from the constant node. Depth D is the maximum logic-node level. LUT6 count and LUT depth are secondary mapping diagnostics. I is a structural graph metric and is not a physical inverter-cell count. We count each graph connection separately, including multiple fanouts from one source; no fanout-dependent cell or wire cost is inferred. Reporting G and I as separate objectives avoids choosing a technology-dependent exchange rate between a logic node and a complement.

III. TWO-STAGE MMIG OPTIMIZATION

The mixed flow builds on existing MOCKTURTLE rewriting, resubstitution, cut rewriting, and refactoring machinery [4], [5]. Stage 1 seeds MIN polarities, runs mixed algebraic and polarity operations, applies bounded window resynthesis, and executes two advanced `compress2rs` rounds. The round objective orders gate count before native complemented-edge count; depth may increase. Balancing is disabled. Candidate passes are guarded by internal equivalence checks. Minority seeding, cone polarity changes, and mixed phase choices are specific to the extended representation; generic rewriting and resubstitution primitives are not new algorithms here. The internal advanced-round

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¹Source and reproducibility artifact: <https://github.com/Mrunal321/mmig-polarity-optimization>, release v1.1-esl-revision.

Algorithm 1 Two-stage native mMIG optimization**Require:** Starting MIG N ; reference function F **Ensure:** Equivalent mixed network N^*

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1:  $X \leftarrow \text{ImportMixed}(N)$ 
2:  $X \leftarrow \text{SeedMinority}(X)$ 
3:  $X \leftarrow \text{GuardedRewriteAndPolarity}(X)$ 
4:  $X \leftarrow \text{GuardedWindowResynthesis}(X)$ 
5: for  $r = 1, 2$  do
6:    $X \leftarrow \text{GuardedCompress2rsRound}(X)$ 
7: end for
8:  $Y \leftarrow \text{ReimportBLIF}(X)$  ▷ MIN labels lost
9:  $Y \leftarrow \text{MixedPhase}(Y)$ 
10:  $N^* \leftarrow \text{TwoPhase}(Y)$  ▷ topology fixed
11: assert  $\text{ABC\_CEC}(F, N^*)$ 
12: return  $N^*$ 

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guard uses the raw native complement count, including constant-source connections, whereas the reported I excludes them. A gate reduction may be accepted with more raw complements within the configured allowance. This distinction matters when interpreting intermediate passes: the paper reports the final nonconstant count, not the optimizer’s internal acceptance score.

Stage 1 writes a BLIF that Stage 2 reimports. This boundary preserves the audited gate-incidence topology but does not preserve explicit MIN labels: all imported gates initially have MAJ labels. Stage 2 applies mixed-phase and two-phase optimization, allowing MAJ/MIN relabeling and input/output phase reassignment without changing that topology. The final native MIN counts therefore arise from Stage 2, not from direct carry-through of Stage 1 type labels. The topology audit finds unchanged gate count, depth, and incidence for all twelve circuits. Algorithm 1 summarizes the executed sequence. The reimport makes the second stage a phase search on the Stage 1 topology rather than a continuation of its in-memory MIN labels. This implementation detail is necessary to reproduce the archived outputs.

The frozen Stage 1 uses three algebraic iterations, four-input window cuts, at most eight internal window nodes, 2,048 expressions, eight matches per root, and 128 roots. It enables don’t-care and zero-gain operations. Exact commands and the hashed binary are in the companion artifact. The second stage only changes phases and node types. The source uses miter-based equivalence guards around the mixed structural passes and checks selected resubstitution candidates. Final raw ABC CEC is an independent verification step against the original benchmark function, not the criterion used to rank candidate graphs.

IV. EXPERIMENTAL EVALUATION

A. Setup and controls

We evaluate twelve EPFL combinational benchmarks [6]. For each benchmark, three fixed flows use the *same* archived starting MIG: T is our local Testa-style fixed-topology pure-MIG polarity control; P is a pure-MIG structural-plus-phase control; and M is the frozen two-stage **zg2**

mMIG flow. P uses a generic standard pre-pass, two **compress2rs** rounds, and a pure-MIG phase pass. It matches the broad structural opportunities of M , including rewriting, resubstitution, cut rewriting, and refactoring, but their libraries, candidate limits, and internal CEC schedules are not identical. Thus $M - P$ is the primary fixed-flow comparison, not a same-runtime or single-feature MIN ablation. Both P and M use area mode, two compress-style rounds, zero-gain moves, a four-input cut setting where available, and a final phase optimization. P receives an additional generic standard pre-pass. The mixed flow alone has minority seeding, cone polarity changes, and its bounded mixed window search. The underlying rewrite libraries and phase search spaces differ, so the comparison controls starting point and broad optimization opportunity, not every primitive operation.

All twelve starting MIGs and all 36 final $T/P/M$ outputs passed direct raw ABC CEC against their original EPFL BLIFs. Each output was mapped by the same ABC script, **strash; if -K 6 -a**. The M profile was developed with knowledge of these twelve benchmarks; the starting MIG choice was also informed by earlier mMIG-score selection. These rows are development-set results, not a holdout generalization study. T is a local implementation of Testa-style transformations, not the released program of Testa *et al.* [2]. The archived starting networks were selected before this control study using earlier mixed-flow results. All three flows share each selected input, but this choice can favor M at the dataset level. We therefore report a fixed-profile observation on the twelve circuits, without claiming an unbiased sample of starting MIGs.

B. Matched pure-MIG comparison

Table I reports the native structural results. Relative to P , M changes aggregate G from 15,205 to 15,070 (−0.89%) and I from 6,501 to 6,021 (−7.38%). Complemented connections fall on eleven circuits and tie on **adder**; gate count falls on nine, ties on two, and rises on **bar**. Fig. 1 shows the edge changes without removing the tie. Relative to T , M has 2.48% fewer gates and 10.79% fewer native complemented connections, but the matched P flow itself improves on T by 1.61% and 3.67%, respectively. Table I lists T ’s edge count per circuit; its fixed-topology pass leaves gate count unchanged. M uses fewer edges than T on ten circuits, ties on **adder**, and is worse on **bar** (624 versus 486). The T comparison alone would overstate the mixed-flow advantage attributable to representation. Published Testa numbers use different starting MIGs and cannot be inserted as same-input results [2]. The largest percentage change is **dec** (16 to 8 edges), where the denominator is small. On the much larger **voter**, the corresponding change is 2,482 to 2,425 edges (−2.30%). The aggregate reduction sums actual connections across the circuit set; it is not the mean of per-circuit percentages. On **bar**, M uses four more logic nodes but 51 fewer complemented connections than P . These circuit-level differences remain visible in Table I.

Table II gives mapping and runtime context. P maps to 5,815 LUT6s and M to 5,867 under the common mapper.

TABLE I

SHARED-INPUT RESULTS. T : LOCAL TESTA-STYLE PHASE CONTROL (NOT THE ORIGINAL PROGRAM); P : MATCHED PURE MIG; M : NATIVE MMIG. G : MAJ+MIN; I : NONCONSTANT COMPLEMENTED EDGES. CHANGES COMPARE M WITH P .

Circuit	$P : G$	$M : G$	ΔG (%)	$T : I$	$P : I$	$M : I$	ΔI (%)	MIN
adder	384	384	0.00	255	255	255	0.00	0
arbiter	3632	3597	-0.96	1079	1026	987	-3.80	205
bar	2460	2464	+0.16	486	675	624	-7.56	45
cavlc	527	524	-0.57	253	243	180	-25.93	57
ctrl	65	64	-1.54	29	29	27	-6.90	1
dec	304	304	0.00	16	16	8	-50.00	4
i2c	848	842	-0.71	409	401	282	-29.68	129
int2float	169	167	-1.18	76	79	54	-31.65	25
max	2093	2052	-1.96	1153	1102	1013	-8.08	70
priority	369	356	-3.52	159	150	129	-14.00	14
router	136	134	-1.47	45	43	37	-13.95	10
voter	4218	4182	-0.85	2789	2482	2425	-2.30	56
Sum	15205	15070	-0.89	6749	6501	6021	-7.38	—

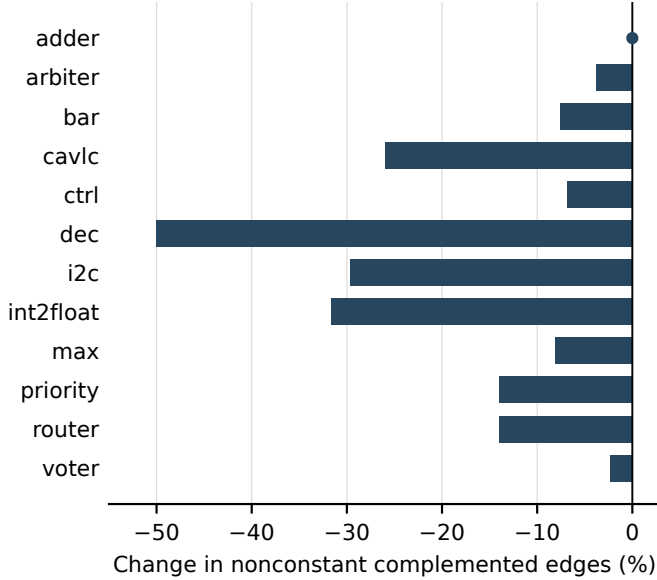


Fig. 1. Change in nonconstant complemented connections for M versus P . Negative values favor the mixed graph; **adder** ties.

TABLE II

AGGREGATE CONTEXT FOR THE THREE FIXED FLOWS.

Flow	G	I	LUT6	Time (s)
T	15454	6749	5863	0.33
P	15205	6501	5815	24.08
M	15070	6021	5867	239.87

The structural polarity reduction does not produce an aggregate LUT6 reduction. M also takes 239.87 s versus 24.08 s for P . The area-oriented flow does not enforce depth non-regression: M is deeper on three circuits, equal on six, and shallower on three; per-circuit depths and LUT depths remain in the artifact. The LUT6 result is a structural mapping diagnostic. It is not an estimate of standard-cell area or of the cost of realizing MIN in a specific technology. The common mapper can absorb phases and restructure logic, so it need not preserve the ordering of native graph-

TABLE III

DIRECT MIN-LOWERING ABLATION ACROSS TWELVE CIRCUITS. THE THREE M ROWS SHARE GATE-INCIDENCE TOPOLOGY.

Representation	G	I	MIN
M native	15070	6021	616
M MAJ-only raw	15070	7028	0
M MAJ-only + phase	15070	6668	0
P pure MIG	15205	6501	0

edge counts.

C. Representation and runtime controls

To test the role of explicit MIN labels, we directly expand the final native MIN nodes into MAJ plus required phases and then apply the same pure-MIG phase optimization. The resulting MAJ-only networks have 6,668 nonconstant complemented connections, versus 6,021 natively and 6,501 for P . The lowered result is 2.57% *higher* than P ; every lowered output passes raw ABC CEC. The reduction is specific to the native MAJ/MIN representation and does not establish a superior MAJ/INV-only realization. Raw lowering before the pure-MIG phase pass has 7,028 connections; the phase pass removes 360, but does not reach P 's 6,501. Gate count, depth, and incidence topology are unchanged by the direct lowering and phase experiment. This ablation isolates representation of the final M graph; it does not prove which Stage 1 pass caused the topology chosen by the mixed flow.

A separate pure-MIG search extends P within an approximate per-circuit M time budget, subject to a 20-round cap. It reaches 15,136 gates, 6,504 complemented connections, and 5,829 LUT6s. Against this control, M has 0.44% fewer gates and 7.43% fewer native complemented connections. Ten circuits reach or slightly exceed their time target; **max** and **voter** stop at the round cap first. This is a sensitivity check, not an equal-runtime result. In particular, the capped P search uses 28.18 s against the 72.35 s M target on **max**, and 50.13 s against 86.55 s on **voter**. Allowing more pure-MIG rounds could change the comparison. Conversely, reducing M 's search or equivalence-checking effort might

change its result. The primary table holds each flow at its frozen profile and makes no equal-work attribution.

D. Scope of the structural result

The three controls answer different questions. T asks how far a local pure-MIG phase pass moves complements without modifying topology. P asks what additional generic pure-MIG structure search provides from that same starting point. M asks what the frozen mixed flow produces after both structural and phase stages. Their common inputs and CEC checks make the output metrics directly comparable, but the flows do not execute the same candidate sequence. In particular, P has no operation identical to the mixed minority-seeding and node-type phase choices, while its generic local libraries are not the same as the mixed implementations. The $M - P$ difference therefore cannot be assigned solely to explicit MIN nodes.

The native-to-lowered comparison provides a narrower representation test. Each lowered circuit preserves the final mixed graph's incidence topology and logic-node count, changing only the representation of MIN and then the allowed pure-MIG phases. Its 647 additional nonconstant complemented connections after pure-MIG phase optimization show that the native MIN labels carry a measurable edge-count benefit for these selected topologies. This benefit says nothing about the cost of a physical MIN implementation. It also does not show that the mixed structural passes would beat a pure-MIG topology search given identical runtime and candidate opportunities.

The twelve benchmarks were used during profile development, and the one chosen starting MIG per benchmark was selected from earlier experiments. We report exact outcomes for those selected cases, not an estimate of expected improvement on unseen circuits. No standard-cell mapping, device model, or post-layout analysis is used to translate I into circuit area or energy. Future evaluation must hold the search budget and target technology fixed before making either claim.

V. RELATED WORK

MIG algebra and Boolean optimization were established by Amarù *et al.* [1]. Testa *et al.* studied fixed-topology

inversion optimization in pure MIGs [2]. Generic rewriting, resubstitution, and refactoring flows provide the structural machinery used by both controls [5]. Aalam *et al.* introduced mMIG synthesis and inversion optimization [3]; our earlier mMIG work studied additional algebra and circuit metrics [7]. This letter focuses on the matched pure-MIG control and direct native-MIN lowering needed to interpret the structural result.

VI. CONCLUSION

Across twelve CEC-verified EPFL circuits, the frozen mixed flow uses 0.89% fewer MAJ+MIN nodes and 7.38% fewer nonconstant complemented graph connections than a matched pure-MIG structural and phase flow. The native edge advantage disappears after direct MIN lowering, while the mixed flow takes about ten times longer and maps to slightly more LUT6s. These results support a representation-specific structural polarity reduction under the tested profiles. They do not show mapped inverter, area, energy, or equal-runtime improvement.

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